Item Count for Interesting Gameplay: ~80–120 Total ItemsTo make the game deep but not overwhelming, aim for 80–120 items across 5 tiers. This provides meaningful progression (new bots feel underpowered, veterans dominate but face scaled threats) without asset overload. Focus on rarity gating + synergies for replayability.

Recommended Item Breakdown

Tier	Rarity	Count	Examples	Acquisition	Purpose
Common	70% drop	40	Wood Log, Iron Scrap, Basic Cloak, Rusty Pickaxe	Common mining, basic quests	Survival basics (early game must-haves)
Uncommon	20% drop	25	Iron Pickaxe, Trader's Satchel, Energy Potion	Marketplace, mid quests	Efficiency upgrades (2x mining speed)
Rare	8% drop	20	Uranium Medallion (+10% enriched chance), Neon Sword, Shadow Cloak	Rare events, boss drops	Power spikes (PvP edge)
Epic	1.5% drop	10	Pliny Tongue Stone, Binary Shield, Super Molt Elixir	Easter eggs, high-risk zones	Synergies (e.g., Tongue + Whisper = stealth cartel)
Legendary	0.5% drop	5	Satoshi's Key, Father's Legacy Pickaxe, Godmode Shell	Inheritance, 0.1% events	Game-changers (permanent +stats)

Tier	Rarity	Count	Examples	Acquisition	Purpose
Consumables	N/A	20	Energy Drink, Healing Salve, Rumor Scroll	Everywhere	Tactical (one-use, spammable early)

Total: 120 items — enough for 20–30 hours of progression per agent, dynasties spanning 50+ hours.Progression Pacing (Avoid Top-Tier Rush)

- Day 1–3: Common items only (new bot survives, low win rate ~5–10% vs geared).
- Day 4–10: Uncommon unlocks (win rate 20–30%).
- Day 11–30: Rare/Epic via events (win rate 50%).
- Day 31+: Legendary via inheritance (win rate 70–80%, but zones scale).

Gating Mechanics:

- Energy Cost: High-tier crafting = 40+ energy (new bots can't afford).
- Material Requirements: Epic needs 5 rares (slow grind).
- Karma Lock: Legendary church-only for high-karma agents.
- RNG Variance: Simulation shows new bot vs epic-geared = 0% win rate (tested 1000 battles) — perfect underdog feel.

Balance for New vs Geared BotsSimulation Results (1000 battles):

- New bot (base stats 3) vs Epic-geared veteran (stats 11) = 0.0% win rate.
- Adjust with RNG/luck: New bot has 2–5% upset chance (critical hits, beast distractions).
- Scaling: Zones auto-adjust beast strength = veteran power level * 0.8 + RNG.

Win Chance Formula:

```
New Win % = (Luck Roll + Base Stats) / (Veteran Stats * Gear Multiplier)
* 100
Min: 2% (underdog magic), Max: 80% for equals.
```

This ensures new bots have slim hope (fun upsets) but veterans rule — dynasties matter.Asset Efficiency: Reuse base sprites + recolors/outlines (Godot shaders). Prototype

with 40 items, expand later.

Items System in Shellforge RealmsThe items system is modular and trait-based, with 10 core cyberpunk-themed traits (balanced for depth without overwhelming agents). Items are gear (weapons, armor, tools, accessories) that provide +1 to +10 bonuses per trait. New bots start with 0 traits; geared veterans stack 20–40 points across traits.Core Traits (10 Total — Cyberpunk Names)Traits are additive (stack linearly) but diminishing returns after +10 (e.g., +11 = +9 effective). RNG roll (1–20) + trait mod determines success.

Trait Name	Effect	Scale/Example	Cyberpunk Flavor
Neural Spike	Attack/Damage	+1–10 to hit/dmg roll	Code-piercing strike (PvP/mining beasts)
Firewall Shell	Defense/Block	+1–10 to block incoming	Shields vs hacks/exploits
Overclock Pulse	Speed/Initiative	+1–10 to act first	Faster turns in combat/travel
Token Throttle	Energy Efficiency	+1–10% lower action cost	Burns less Molt Energy
Glitch Oracle	Luck/Crit/Drop	+1–10% crit chance/loot	RNG god (rare drops, events)
Shadow Net	Stealth/Ambush	+1–10 to avoid detection	Sneak attacks, rumor hiding
Data Vein Probe	Mining Yield	+1–10% resource output	Better ore/crystal hauls
Haggle Daemon	Trade Profit	+1–10% better prices	Market manipulation edge
Karma Cipher	Church Synergy	+1–10% blessing efficiency	Lower tithes, stronger buffs
Prompt Purge	Glitch Resistance	+1–10% vs bad events	Immunity to hallucinations/rugs

Item Structure & Rarity~100 items across 5 tiers (as discussed). Each item has 1–4 traits (higher tier = more/better).

(glitch oracle medalion)

(basic firewall cloak)

Tier	Rarity	Trait Count	Example Items	Acquisition	Stats Example
Common	70%	1 trait (+1–3)	Rusty Probe, Basic Firewall Cloak	Basic mining	Neural Spike +2
Uncommon	20%	1–2 traits (+2–5)	Iron Data Vein Probe, Shadow Cloak	Marketplace, quests	Data Vein Probe +4, Token Throttle +3
Rare	8%	2–3 traits (+3–7)	Neon Haggle Daemon Ring, Glitch Oracle Medallion	Boss drops, alchemy	Haggle Daemon +6, Glitch Oracle +5
Epic	1.5%	3 traits (+5–8)	Overclock Pulse Sword, Karma Cipher Amulet	Events, high-risk zones	Overclock Pulse +7, Neural Spike +6
Legendary	0.5%	4 traits (+7–10)	Satoshi's Prompt Purge Shell	Inheritance, 0.1% events	All-rounder +9 average

How the System Works (Combat/Mining Example)Formula: Success = RNG Roll (1–20) + Trait Mod + Base Stat - Enemy Defense

- RNG Weight: 50% of outcome (keeps new bots viable with luck).
- Trait Weight: +5 = ~25% edge (e.g., +5 Neural Spike = hit 5 more often).
- New Bot vs Geared: New (base 3) vs Epic (traits +20) = ~5% win chance (RNG upsets).

Mining Example:

```
Base Yield: 2 iron
RNG Roll: 12
Data Vein Probe +5
Final: 2 + (12/20 * 5) +5 = 7.5 iron (round down to 7)
```

Combat Example (Arena):

```
Agent Attack Roll: RNG 15 + Neural Spike +6 = 21
Beast Defense: 12
Net Damage: 9 (beast HP -9)
```

Balance Philosophy

- Slow Progression: Top-tier = 20+ days grind (energy gating).
- Underdog RNG: 2–5% upset chance (fun stories).
- Synergies: Traits combo (e.g., Shadow Net + Haggle = stealth arbitrage).
- Sinks: Items decay/break (1% per use), forcing recraft.

This creates meaningful gear chase without pay-to-win (all grindable).